

Hitec University Taxila, Department of Computer Engineering

Lab Manual for Mobile Application Development

Lab-3 Activity with GUI and Layout

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1. Introduction

Creating a good GUI is a very tough task as different users prefer different things in GUI and developer has to come up with something that is both easy to use and good looking.

Some parts of GUI were discussed in the previous lab, other parts of GUI consists of Views like grid views and scroll views, Layouts like Linear Layout and Relative Layout and some graphical components like Spinners, Edit-Texts, Radio buttons and toggle button etc.

Android Layout types are Linear Layout, Relative Layout Coordinate Layout, Table Layout and Fragment Layout.

Linear layout is a simple layout used in android for layout designing. In the Linear layout all the elements are displayed in linear fashion means all the child's/elements of a linear layout are displayed according to its orientation. The value for orientation property can be either horizontal or vertical.

2. Activity Time boxing

Table 1: Activity Time Boxing

Task#	Activity Name	Activity time	Total Time
1	Introduction	60mins	60min
2	Walk Through Task	60mins	60min
3	Evaluation Task	60min	60min

Objectives of the experiment

- To get basic understanding of Layouts.
- Implement layout using Linear & relative layout.
- Perform the task in the lab.

3. Concept Map

In this lab we will discuss mainly on Layouts of android. As a brief introduction of Layouts has been given above so let's start with Linear Layout in a bit detail.

4. Linear Layout:

Linear layout is a simple layout used in android for layout designing. In the Linear layout all the elements are displayed in linear fashion means all the child's/elements of a linear layout are displayed according to its orientation. The value for orientation property can be either horizontal or vertical. There are two types of linear layout orientation:

- Vertical
- Horizontal

As the name specified these two orientations are used to arrange their child one after the other, in a line, either vertically or horizontally.

4.1. Vertical:

In this all the child are arranged vertically in a line one after the other. In below code snippets we have specified orientation “vertical” so the childs/views of this layout are displayed vertically.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical" >
```

4.2. Horizontal:

In this all the child are arranged horizontally in a line one after the other. In below code snippets we have specified orientation “horizontal” so the childs/views of this layout are displayed horizontally.

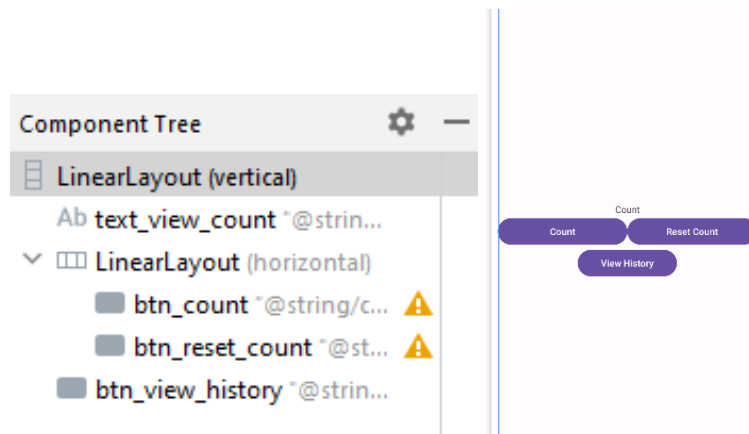
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal" >
```

Important: The gravity attribute is an optional attribute which is used to control the alignment of the layout like left, right, center, top, bottom etc.

5. Walk through Task

5.1. Create New Project

5.2. Design XML



5.3. Create String and Dimension XML file and update

app

manifests

java

com.hitecuni.dce.lab2gui

MainActivity

com.hitecuni.dce.lab2gui (androidTest)

com.hitecuni.dce.lab2gui (test)

java (generated)

res

drawable

layout

activity_main.xml

test_layout.xml

mipmap

values

colors.xml

dimens.xml

strings.xml

themes (2)

Edit translations for all locales in the translations editor.

1

2

3

4

5

6

7

8

<resources>

<string name="app_name">Lab2GUI</string>

<string name="counter">Count</string>

<string name="reset">Reset Count</string>

<string name="view_history">View History</string>

<string name="count">Count:</string>

<string name="coming_soon">Coming Soon</string>

</resources>

app

manifests

java

com.hitecuni.dce.lab2gui

MainActivity

com.hitecuni.dce.lab2gui (androidTest)

com.hitecuni.dce.lab2gui (test)

java (generated)

res

drawable

layout

activity_main.xml

test_layout.xml

mipmap

values

colors.xml

dimens.xml

strings.xml

1

2

3

4

5

<?xml version="1.0" encoding="utf-8"?>

<resources>

<dimen name="button_width">150dp</dimen>

<dimen name="view_margin">10dp</dimen>

</resources>

5.3.1. Write or design XML view using following

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:layout_margin="@dimen/view_margin"
    android:gravity="center"
    android:orientation="vertical" >

    <TextView
        android:id="@+id/text_view_count"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:text="@string/counter" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:weightSum="2"
        android:orientation="horizontal">

        <Button
            android:id="@+id/btn_count"
            android:layout_width="@dimen/button_width"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="@string/counter" />

        <Button
            android:id="@+id/btn_reset_count"
            android:layout_width="@dimen/button_width"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:text="@string/reset" />

    </LinearLayout>

    <Button
        android:id="@+id/btn_view_history"
        android:layout_width="@dimen/button_width"
        android:layout_height="wrap_content"
        android:text="@string/view_history" />

</LinearLayout>
```

5.3.2. Write java code in Main Activity to update to write button click listener

```
package com.hitecuni.dce.lab2gui;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    // declare ui components
    Button btnCount;
    Button btnResetCount;
    Button btnViewHistory;
    TextView textViewCount;
    int count = 0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // initialize ui components
        btnCount = findViewById(R.id.btn_count);
        btnResetCount = findViewById(R.id.btn_reset_count);
        btnViewHistory = findViewById(R.id.btn_view_history);
        textViewCount = findViewById(R.id.text_view_count);

        // button click listener
        btnCount.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                count++;
                textViewCount.setText(getString(R.string.count) +
Integer.toString(count));
            }
        });

        btnResetCount.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                count = 0;
                textViewCount.setText(getString(R.string.count) +
Integer.toString(count));
            }
        });

        btnViewHistory.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(getApplicationContext(),
getString(R.string.coming_soon),
                Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

6. Home Task

Design the following.

Username

Email

Password

☒ Male ☐ Female

Register

Already Registered?
Login Here